

Problem Description

A and B are playing a game. In this game, both of them are initially provided with a **list of n numbers**. (Both have the same list but their own copy).

Now, they both have a different strategy to play the game. A picks the element from **start of his list**. B picks from the **end of his list**.

You need to generate the result in form of an output list.

Method to be followed at each step to build the output list is:

1. If the number picked by A is **bigger than** B then this step's **output is 1** . B **removes** the number that was picked from their list.
2. If the number picked by A is **smaller than** B then this step's **output is 2** . A **removes** the number that was picked from their list.
3. If both have the **same number** then this step's **output is 0** . **Both A and B remove** the number that was picked from their list.

This game **ends** when at least one of them has no more elements to be picked i.e. when the **list gets empty**.

Output the built output list.

Constraints

$$1 \leq N \leq 10^6$$

$$1 \leq \text{num} \leq 10^9$$

Input format:

First line consists of a number n , size of the list provided.

Next line consists of n numbers separated by space.

Output format:

Output the required output list.

Logical Test Cases

Test Case 1

INPUT (STDIN)

3
1 2 3

EXPECTED OUTPUT

2 2 0

Test Case 2

INPUT (STDIN)

5
5 7 1 2 3

EXPECTED OUTPUT

1 1 1 2 0 2 2 2

Mandatory Test Cases

Test Case 1

KEYWORD

if(a[i]>a[j])

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

6

Test Case 2

TOKEN COUNT

160

Test Case 3

NLOC

36